

A REARRANGEMENT INEQUALITY FOR THE DISCRETE HILBERT TRANSFORM, WITH AN APPLICATION TO OPTIMAL CGLMP MEASUREMENTS

ANSH MISHRA AND ARYAN SENTHILKUMAR

ABSTRACT. For disjoint subsets A, B of the cyclic group $\mathbb{Z}_N$ of prescribed sizes we show that the pairing $\sum_{x \in A, y \in B} \cot(\pi(x - y)/N)$, the bilinear form of the discrete Hilbert transform, is largest exactly when A and B are adjacent intervals with B immediately before A . The proof has three ingredients: an exact exchange identity, which forces every maximiser to be cyclically ordered; a continuous version of the inequality, based on the circle analogue of a theorem of Stein and Weiss (the distribution of the conjugate function of an indicator function depends only on the measure of the set); and an explicit estimate of the discretisation error at the junctions between blocks, which excludes cyclically ordered configurations that wind around the circle more than once.

We apply the inequality to Open Quantum Problem 27B of IQOQI Vienna, which asks to show that the measurements of Durt, Kaszlikowski and Żukowski (DKZ) are necessarily the optimal ones for the CGLMP_d Bell inequality on maximally entangled states. The optimality statement is equivalent to an inequality for d -tuples of unitaries of order four (a $\mathbb{Z}_4$ clock model). We prove that inequality for every d when the unitaries commute. In physical terms: among maximally entangled strategies whose four link operators coincide, the DKZ measurements are optimal and, up to local unitaries, unique, in every dimension d . The general (non-commuting) case remains open for $d \geq 13$; for $d \leq 12$ it was settled earlier by exact computer-assisted certificates.

1. INTRODUCTION

1.1. The problem. The CGLMP inequalities of Collins, Gisin, Linden, Massar and Popescu [5] are Bell inequalities for two parties with two measurement settings each and d outcomes per measurement. Write A_1, A_2 and B_1, B_2 for the outcomes (elements of $\mathbb{Z}_d$) of Alice's and Bob's measurements. The CGLMP expression is

$$(1) \quad I_d = \sum_{k=0}^{\lfloor d/2 \rfloor - 1} \left(1 - \frac{2k}{d-1}\right) \left([P(A_1 = B_1 + k) + P(B_1 = A_2 + k + 1) + P(A_2 = B_2 + k) + P(B_2 = A_1 + k)] - [P(A_1 = B_1 - k - 1) + P(B_1 = A_2 - k) + P(A_2 = B_2 - k - 1) + P(B_2 = A_1 - k - 1)] \right),$$

where $P(A_x = B_y + k)$ is the probability that $A_x - B_y \equiv k \pmod{d}$ when the settings x and y are used.¹ Local hidden variable models satisfy $I_d \leq 2$ [5]. On the maximally entangled state of two d -level systems, the measurements found numerically by Kaszlikowski et al. [12] and described analytically by Durt, Kaszlikowski and Żukowski [7], namely diagonal phase shifts followed by

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¹This is the reading under which the local bound is derived in [5, eqs. (7)–(11)] and under which the measurements of [5, eqs. (12)–(15)] give the value (2); see check (R7) in Section 5.

a discrete Fourier transform, which we call the *DKZ measurements*, give the value

$$(2) \quad I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d},$$

for example $I_{\text{ME}}(3) = 2.8729\dots$ and $I_{\text{ME}}(4) = 2.8962\dots$. Numerical searches [12, 7, 5] found no better measurements on the maximally entangled state, while for $d \geq 3$ the maximally entangled state is not the optimal state for I_d [1]. Part B of Open Quantum Problem 27 of the IQOQI Vienna list (“The power of CGLMP inequalities”, [10]) records that the observables which numerically maximise the CGLMP violation on a maximally entangled state are measurements in the computational basis transformed by a discrete Fourier transform and diagonal unitaries, and asks to show that this is necessarily the case. It also asks to show that these measurements are optimal for noise resistance and for the Kullback–Leibler distance to local models; we do not address those two parts here.

1.2. What was known. For $d = 2$ the CGLMP expression is the CHSH expression and the statement is Tsirelson’s bound [4]. For $d \geq 3$ the evidence was numerical [12, 7, 5]. For $d = 3$, a semidefinite relaxation shows that the maximal value on maximally entangled states of any dimension agrees with the DKZ value to within 10^{-10} [14] (this is the partial result listed in [10]). Ioannou and Rosset [9] obtained sum-of-squares certificates for the maximal quantum value of I_3 and I_4 over all states; by [1] this is a different optimisation problem from the one on maximally entangled states. In [17] we gave exact sum-of-squares certificates, with cyclotomic coefficients and checked by independent exact verifiers, which prove that the DKZ measurements are optimal among projective measurements on maximally entangled states of any local dimension for $d = 3, \dots, 12$, together with a rigidity statement. The SATWAP expressions of Salavrakos et al. [19] are maximally violated by the maximally entangled state with measurements of CGLMP type, admit sum-of-squares decompositions for all d , and underlie self-testing results [11, 20]; they are different Bell expressions and do not bear directly on I_d . Within the family of Fourier measurements with four linear phase offsets, the DKZ point was recently found numerically to be a strict local maximum for $d = 2, \dots, 6$ [22, Sec. V]. To our knowledge no argument valid for all d was available, even within a restricted class of measurements.

1.3. The clock model and its commutative sector. The starting point of this paper is an exact reduction, stated as Theorem 2.3 below and proved in Appendix A (it is also part of the material of [17]). Put

$$\psi_m = \frac{\pi m}{2d}, \quad h_m = \sec \psi_m - i \csc \psi_m \quad (1 \leq m \leq d-1),$$

and for unitaries $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = 1$ let

$$(3) \quad F(V) = \sum_{0 \leq j < k \leq d-1} \operatorname{Re}[h_{k-j} \operatorname{tr}(V_j V_k^*)], \quad \operatorname{tr} = \frac{1}{M} \operatorname{Tr}.$$

For every projective strategy on a maximally entangled state there is such a family with $I_d = 4F(V)/(d(d-1))$, and conversely every such family comes from a strategy. The DKZ strategy corresponds to $V_0 = \dots = V_{d-1} = 1$, with

$$(4) \quad F_{\text{DKZ}} = F(1, \dots, 1) = \sum_{m=1}^{d-1} (d-m) \sec \psi_m, \quad I_{\text{ME}}(d) = \frac{4F_{\text{DKZ}}}{d(d-1)}.$$

So the optimality of the DKZ measurements on maximally entangled states (of any dimension, for projective measurements) is equivalent to the inequality $F(V) \leq F_{\text{DKZ}}$ for all M and all d -tuples of unitaries of order dividing four.

When the V_k commute, every joint eigenvector v carries a configuration $a \in \mathbb{Z}_4^d$ with $V_k v = i^{-a_k} v$, and $F(V)$ is an average of the values

$$(5) \quad F(a) = \sum_{0 \leq j < k \leq d-1} \operatorname{Re}[h_{k-j} i^{a_k - a_j}], \quad a \in \mathbb{Z}_4^d.$$

This is a classical long-range four-state clock model on d sites (equivalently, a long-range Ising ring with antiperiodic boundary conditions, Section 6). The corresponding Bell strategies are genuinely quantum: the DKZ strategy itself belongs to this commutative sector. We call a *one-step* if $a_k = c + [k \geq r]$ for some $c \in \mathbb{Z}_4$ and $r \in \{0, \dots, d-1\}$; there are $4d$ one-step configurations, and $a = 0$ is one of them.

1.4. Results. For $N \geq 2$ put $\kappa(u) = \cot(\pi u/N)$ for $u \in \mathbb{Z}_N \setminus \{0\}$ and $\kappa(0) = 0$, and for $X, Y \subseteq \mathbb{Z}_N$

$$\langle X, Y \rangle = \sum_{x \in X} \sum_{y \in Y} \kappa(x - y).$$

For integers $p \leq q$ let $[p, q] = \{p, p+1, \dots, q-1\}$, read modulo N . The function κ is the kernel of the discrete (periodic) Hilbert transform on $\mathbb{Z}_N$, so $\langle X, Y \rangle$ is its bilinear form on indicator functions.

Theorem A (rearrangement inequality). *Let $N \geq 2$ and let $A, B \subseteq \mathbb{Z}_N$ be disjoint, $|A| = a$, $|B| = b$. Then*

$$\langle A, B \rangle \leq \Phi_N(a, b) := \langle [0, a], [-b, 0] \rangle.$$

If $a, b \geq 1$ and $a + b < N$, equality holds if and only if $(A, B) = (r + [0, a], r + [-b, 0])$ for some $r \in \mathbb{Z}_N$.

In words: the pairing is largest when A and B are intervals and B ends exactly where A begins. If $a = 0$, $b = 0$ or $a + b = N$, both sides vanish (Section 4). The discrete Hilbert transform has mostly been studied through Hilbert-type inequalities and operator norms (see, e.g., [18, 2]); we have not found Theorem A, or an equivalent statement, in the literature. Its continuous analogue (Proposition 4.6) is a direct consequence of known results [21, 13] and the bathtub principle, but the discrete statement does not follow from it (Remarks 4.7 and 4.8).

Theorem B (commutative sector, all d). *Let $d \geq 2$ and let $V_0, \dots, V_{d-1} \in U(M)$ be pairwise commuting with $V_k^4 = 1$. Then $F(V) \leq F_{\text{DKZ}}$. Equality holds if and only if every joint eigenvector v of the V_k satisfies $V_k v = i^{-a_k} v$ with a a one-step configuration a (which may depend on v). The same holds for commuting unitaries of order dividing four in any von Neumann algebra with a faithful normal tracial state τ in place of tr .*

For the physical statement, encode Alice's projective measurements $\{P_a^x\}_{a \in \mathbb{Z}_d}$ and Bob's $\{Q_b^y\}_{b \in \mathbb{Z}_d}$ on $\mathbb{C}^D \otimes \mathbb{C}^D$ by the unitaries $U_x = \sum_a w^a P_a^x$ and $U'_y = \sum_b w^b Q_b^y$, $w = e^{2\pi i/d}$, and put

$$(6) \quad R_1 = U_2, \quad R_2 = (U'_2)^T, \quad R_3 = U_1, \quad R_4 = (U'_1)^T,$$

where the transpose is taken in the Schmidt basis of the maximally entangled state $\Phi_D = D^{-1/2} \sum_{i=1}^D |ii\rangle$. The *links* of the strategy are the following unitaries:

$$L_n^{(0)} = R_1^n R_2^{-n}, \quad L_n^{(1)} = R_2^n R_3^{-n}, \quad L_n^{(2)} = R_3^n R_4^{-n}, \quad L_n^{(3)} = w^{-n} R_4^n R_1^{-n} \quad (1 \leq n \leq d-1);$$

their normalised traces are the Fourier coefficients of the distributions of the outcome differences along the chain A_2, B_2, A_1, B_1 (Section 2). We say that a strategy has *equal links* if $L_n^{(0)} = L_n^{(1)} = L_n^{(2)} = L_n^{(3)}$ for every n .

Corollary C. *Let $d \geq 2$. Every projective strategy on a maximally entangled state Φ_D (any D) with equal links satisfies $I_d \leq I_{\text{ME}}(d)$. Equality holds if and only if d divides D and, for some unitary $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$, the local unitary $u \otimes \bar{u}$ maps the strategy to the DKZ strategy tensored with the identity on $\mathbb{C}^{D/d}$. In particular, each of the 4^d clock-phase strategies of Section 2.5 satisfies $I_d \leq I_{\text{ME}}(d)$, with equality exactly for the $4d$ one-step ones.*

The equal-link strategies are exactly those whose reduced family V in Theorem 2.3 commutes (Proposition 2.4). Theorem B follows from Theorem A through an exact identity (Lemma 3.1): with $N = 4d$ and $S_a = \{k - da_k : 0 \leq k \leq d - 1\} \subseteq \mathbb{Z}_{4d}$,

$$F(a) = \frac{1}{2} \langle S_a, S_a - d \rangle - \frac{d}{2},$$

and S_a is an interval exactly when a is one-step. So Theorem B is the case $N = 4d$, $a = b = d$ of Theorem A.

1.5. Method and difficulty. The proof of Theorem A (Section 4) has four steps. Let $C = \mathbb{Z}_N \setminus (A \cup B)$. (i) The three pairings $\langle A, B \rangle$, $\langle B, C \rangle$, $\langle C, A \rangle$ are equal, and exchanging the colours of an adjacent pair of type (A, B) , (B, C) or (C, A) increases the common value by an explicit positive amount. Hence every maximiser is cyclically ordered: going around the circle the colours run through $w \geq 1$ rounds of blocks C, B, A . The configurations with $w = 1$ are the rotations of $([0, a), [-b, 0))$. (ii) Smearing every point to a unit cell writes $\langle A, B \rangle$ as a continuous pairing plus a sum of explicit junction terms. (iii) The continuous pairing is maximised by adjacent arcs; this follows from the fact that the distribution of the conjugate function of the indicator of a finite union of arcs, on the complement of the set, depends only on the total length (a circle version of results of Stein and Weiss [21] and Laeng [13], related to Boole’s identity [3]), combined with the bathtub principle. (iv) A configuration with w rounds has exactly w junctions $B|A$, each of which costs about $(2 \log 2 - 1)N/\pi$ in the junction terms; this beats all other junction contributions as soon as $w \geq 2$.

Step (iv) cannot be avoided. The continuous problem in step (iii) has many maximisers, including the w -fold covers of a pair of adjacent arcs (Remark 4.7), so the discrete inequality is decided by the discretisation error, which is of order N . In the language of the clock model this is a near-degeneracy between the DKZ configuration and “ w -fold” configurations (Section 6), and it suggests why certificates that are local in the site index fail for large d : for instance, the Sherali–Adams relaxation of the clock model built from all three-site marginals is exact for $d \leq 8$ but not for $9 \leq d \leq 12$ (numerical; Section 6).

1.6. What remains open. Theorem B covers commuting families. Whether $F(V) \leq F_{\text{DKZ}}$ holds for non-commuting families, which is what remains of the optimality statement of Problem 27B on maximally entangled states, is open for $d \geq 13$. In Section 6 we formulate an operator version of Theorem A that would imply it. The cyclic symmetry extends to operators verbatim and the exchange identity only partially; we do not know operator versions of the continuous inequality and of the junction estimate.

1.7. Organisation. Section 2 recalls the reduction to the clock model and characterises the commutative sector. Section 3 proves the cotangent representation and deduces Theorem B and Corollary C from Theorem A. Section 4 proves Theorem A. Section 5 lists numerical cross-checks, which are not part of the proofs, and Section 6 discusses the non-commutative case. Appendix A contains the proof of the reduction.

2. THE CGLMP EXPRESSION AND THE CLOCK MODEL

Throughout, $d \geq 2$, $w = e^{2\pi i/d}$, $z = e^{2\pi i/(4d)}$ (so $z^4 = w$ and $z^d = i$), and $m(t) \in \{0, \dots, d-1\}$ denotes the residue of an integer t modulo d .

2.1. A chain form of the CGLMP expression. Relabel the settings as $X_1 = A_2$, $Y_1 = B_2$, $X_2 = A_1$, $Y_2 = B_1$, and put

$$(7) \quad S = \mathbb{E} m(X_1 - Y_1) + \mathbb{E} m(Y_1 - X_2) + \mathbb{E} m(X_2 - Y_2) + \mathbb{E} m(Y_2 - X_1 - 1),$$

each expectation being taken in the distribution of the pair of outcomes that occurs in it.

Lemma 2.1. *For every behaviour $P(a, b|x, y)$, $I_d = 4 - \frac{2}{d-1} S$.*

Proof. Consider the terms of (1) that involve A_1 and B_1 , and let $t = m(A_1 - B_1)$. For $0 \leq k \leq \lfloor d/2 \rfloor - 1$ the event $t = k$ has weight $1 - \frac{2k}{d-1}$, and the event $t = d-1-k$ has weight $-(1 - \frac{2k}{d-1}) = 1 - \frac{2(d-1-k)}{d-1}$. If d is odd, the remaining value $t = (d-1)/2$ does not occur and $1 - \frac{2t}{d-1} = 0$ there. So these terms equal $\mathbb{E} f(t)$ with $f(t) = 1 - \frac{2t}{d-1}$ for all $t \in \{0, \dots, d-1\}$, that is $1 - \frac{2}{d-1} \mathbb{E} m(A_1 - B_1)$. In the same way the terms with (B_1, A_2) , (A_2, B_2) and (B_2, A_1) give $1 - \frac{2}{d-1} \mathbb{E} m(B_1 - A_2 - 1)$, $1 - \frac{2}{d-1} \mathbb{E} m(A_2 - B_2)$ and $1 - \frac{2}{d-1} \mathbb{E} m(B_2 - A_1)$. In the relabelled variables the sum of the four expectations is S . $\square$

Remark 2.2. The local bound $I_d \leq 2$ reads $S \geq d-1$; this is the form in which the inequality is written in [10], where it is attributed to Gill. For no-signalling behaviours one also has $S = d\Pi - 1$ with $\Pi = P(X_1 < Y_1) + P(Y_1 < X_2) + P(X_2 < Y_2) + P(Y_2 \leq X_1)$ (outcomes compared as integers in $\{0, \dots, d-1\}$), which follows from $m(u-v) = u-v+d[u < v]$ for $u, v \in \{0, \dots, d-1\}$. So $I_d \leq 2$ is equivalent to $\Pi \geq 1$, the form of the CGLMP inequality used by Zohren and Gill [23, eq. (3)]. We do not use these facts.

2.2. Maximally entangled strategies. A projective strategy on Φ_D gives $P(a, b|x, y) = \langle \Phi_D | P_a^x \otimes Q_b^y | \Phi_D \rangle = \tau(P_a^x (Q_b^y)^T)$ with $\tau = \frac{1}{D} \text{Tr}$. With the unitaries (6), which satisfy $R_i^d = 1$, and the Fourier expansion

$$(8) \quad m(t) = \frac{d-1}{2} + \sum_{n=1}^{d-1} c_n w^{nt}, \quad c_n = -\frac{1}{1-w^{-n}},$$

one finds $\mathbb{E} w^{n(X_1-Y_1)} = \tau(R_1^n R_2^{-n})$, $\mathbb{E} w^{n(Y_1-X_2)} = \tau(R_2^n R_3^{-n})$, $\mathbb{E} w^{n(X_2-Y_2)} = \tau(R_3^n R_4^{-n})$ and $\mathbb{E} w^{n(Y_2-X_1-1)} = w^{-n} \tau(R_4^n R_1^{-n})$. Hence

$$(9) \quad S = 2(d-1) + \sum_{n=1}^{d-1} c_n \Lambda_n(R), \quad \Lambda_n(R) = \sum_{i=0}^3 \tau(L_n^{(i)}),$$

with the links $L_n^{(i)}$ defined in the introduction. Conversely, any four unitaries $R_1, \dots, R_4$ on $\mathbb{C}^D$ with $R_i^d = 1$ come from a projective strategy on Φ_D , through their spectral projections. (The coefficient in (8) is $\frac{1}{d} \sum_{t=0}^{d-1} t w^{-nt} = -1/(1-w^{-n})$ for $n \neq 0$.)

2.3. The reduction. For unitaries $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = 1$ let X be the cyclic shift $X|k\rangle = |k+1\rangle$ on $\mathbb{C}^d$ (indices modulo d) and define the *covariant strategy* R^V on $\mathbb{C}^d \otimes \mathbb{C}^M$ by

$$(10) \quad W = \sum_{k=0}^{d-1} |k\rangle\langle k| \otimes z^k V_k, \quad R_1^V = X \otimes \mathbf{1}_M, \quad R_{i+1}^V = W R_i^V W^* \quad (i = 1, 2, 3).$$

Theorem 2.3 (reduction to the clock model). (1) *For every covariant strategy, $S(R^V) = 2(d-1) - \frac{2}{d} F(V)$, that is $I_d(R^V) = \frac{4F(V)}{d(d-1)}$.*

(2) *For every projective strategy R on Φ_D there are M (one can take $M = 4D$) and unitaries $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = 1$ such that R is unitarily equivalent to a direct summand of R^V (simultaneous conjugation of $R_1, \dots, R_4$) and $I_d(R) = I_d(R^V) = \frac{4F(V)}{d(d-1)}$.*

Consequently, $I_d \leq I_{\text{ME}}(d)$ holds for all projective strategies on maximally entangled states of all dimensions if and only if $F(V) \leq F_{\text{DKZ}}$ for all M and all $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = 1$.

The family V in (2) is produced by a twirl over a cyclic symmetry of order $4d$ followed by the Stone–von Neumann theorem for $\mathbb{Z}_d$; it is unique up to simultaneous unitary conjugation of the V_k (Appendix A). We call it the *reduced family* of R .

2.4. The DKZ strategy. Let $W_0 = \text{diag}(z^k)_{k=0}^{d-1}$, so that $R^{(0)} := R^V$ for $M = 1$, $V \equiv 1$, is $R_i^{(0)} = W_0^{i-1} X W_0^{-(i-1)}$. The measurement procedure of [5, eqs. (12)–(15)] (diagonal phases $e^{2\pi i \alpha_x j/d}$ for Alice and $e^{2\pi i \beta_y j/d}$ for Bob, then a discrete Fourier transform on each side, then a measurement in the computational basis, with $\alpha = (0, \frac{1}{2})$ and $\beta = (\frac{1}{4}, -\frac{1}{4})$) amounts to measuring in the orthonormal bases $|k\rangle_{A,x} = d^{-1/2} \sum_j e^{-2\pi i j(k+\alpha_x)/d} |j\rangle$ and $|l\rangle_{B,y} = d^{-1/2} \sum_j e^{2\pi i j(l-\beta_y)/d} |j\rangle$; this gives the joint probabilities $P(A_x = k, B_y = l) = (2d^3 \sin^2(\pi(k - l + \alpha_x + \beta_y)/d))^{-1}$ of [5, eq. (15)]. Writing $D_\gamma = \text{diag}(w^{\gamma j})$ and $F|j\rangle = d^{-1/2} \sum_k w^{jk} |k\rangle$ one has $U_x = D_{\alpha_x}^* (F^* Z_0 F) D_{\alpha_x} = D_{\alpha_x}^* X D_{\alpha_x}$ with $Z_0 = \text{diag}(w^k)$, and similarly $(U_y')^T = D_{\beta_y} X D_{\beta_y}^*$. Since $D_{1/4} = W_0$, the chain (6) of the DKZ strategy is

$$(R_1, R_2, R_3, R_4) = (W_0^{-2} X W_0^2, W_0^{-1} X W_0, X, W_0 X W_0^{-1}) = G R^{(0)} G^*, \quad G = W_0^{-2}.$$

Thus DKZ is, up to the local unitary $G \otimes \bar{G}$ (which fixes Φ_d), the covariant strategy with $V \equiv 1$, and $I_d(\text{DKZ}) = 4F_{\text{DKZ}}/(d(d-1)) = I_{\text{ME}}(d)$ by Theorem 2.3(1).

2.5. The commutative sector. For $a \in \mathbb{Z}_4^d$ the *clock-phase strategy* R^a on Φ_d is the covariant strategy with $M = 1$ and $V_k = i^{-a_k}$, that is

$$R_1^a = X, \quad R_{i+1}^a = W_a R_i^a W_a^*, \quad W_a = \text{diag}(z^k i^{-a_k})_{k=0}^{d-1}.$$

All four of its measurement bases are Fourier bases dressed by diagonal phases, consecutive ones differing by the same diagonal unitary W_a , and $I_d(R^a) = 4F(a)/(d(d-1))$ with $F(a)$ as in (5). If $V_0, \dots, V_{d-1}$ commute, a common eigenbasis (v_μ) splits R^V into the direct sum $\bigoplus_\mu R^{a(\mu)}$, where $V_k v_\mu = i^{-a_k(\mu)} v_\mu$.

Proposition 2.4 (equal links). *A projective strategy on Φ_D has equal links if and only if its reduced family consists of pairwise commuting unitaries. In particular every clock-phase strategy, and every covariant strategy with commuting V , has equal links.*

The proof is in Appendix A. The equal-link condition for $n = 1$ is one of the conditions that appear in the rigidity analysis of [17].

3. THE COTANGENT REPRESENTATION; PROOFS OF THEOREM B AND COROLLARY C

Lemma 3.1 (cotangent representation). *Let $N = 4d$ and $\kappa(u) = \cot(\pi u/N)$ as above. For $a \in \mathbb{Z}_4^d$ put $S_a = \{k - da_k \bmod N : 0 \leq k \leq d-1\}$. Then S_a contains exactly one element of each residue class modulo d , every such transversal is S_a for exactly one a , the sets S_a and $S_a - d$ are disjoint, and*

$$F(a) = \frac{1}{2} \sum_{\substack{x, y \in S_a \\ x \neq y}} \sec \frac{\pi(x-y)}{2d} = \frac{1}{2} \langle S_a, S_a - d \rangle - \frac{d}{2}.$$

Moreover S_a is an interval $[c, c+d)$ if and only if a is one-step, and then $(S_a, S_a - d) = ([c, c+d), [c-d, c))$.

Proof. Let $x_k = k - da_k$. For $j < k$, $m = k - j$ and $s = a_k - a_j$ we have $x_k - x_j = m - ds$ and $\operatorname{Re}[h_m i^s] = \sec(\psi_m - s\pi/2)$ (for $s = 0, 1, 2, 3$ both sides equal $\sec \psi_m, \csc \psi_m, -\sec \psi_m, -\csc \psi_m$), which is $\sec(\pi(x_k - x_j)/(2d))$; the function $u \mapsto \sec(\pi u/(2d))$ is even and has period $4d$. This gives the first expression for $F(a)$. The first assertions are clear since $x_k \equiv k \pmod{d}$ and a_k ranges over $\mathbb{Z}_4$.

Next, $\cot p - \cot q = \sin(q - p)/(\sin p \sin q)$ and $\sin(\frac{t}{2} + \frac{\pi}{4}) \sin(\frac{t}{2} - \frac{\pi}{4}) = -\frac{1}{2} \cos t$ give $\sec t = \frac{1}{2}[\cot(\frac{t}{2} + \frac{\pi}{4}) - \cot(\frac{t}{2} - \frac{\pi}{4})]$. With $t = 2\pi(x - y)/N$ and $\pi/4 = \pi d/N$ this reads

$$\sec \frac{\pi(x - y)}{2d} = \frac{1}{2}[\kappa(x - y + d) - \kappa(x - y - d)] \quad (x \not\equiv y \pmod{d}),$$

where no pole occurs because $x - y \pm d \not\equiv 0 \pmod{4d}$. Summing over ordered pairs $x \neq y$ in S_a and using that κ is odd, the two halves are equal, so $F(a) = \frac{1}{2} \sum_{x \neq y} \kappa(x - y + d) = \frac{1}{2}[\langle S_a, S_a - d \rangle - d\kappa(d)]$ and $\kappa(d) = \cot(\pi/4) = 1$. If $x = y - d$ with $x, y \in S_a$, then $x \equiv y \pmod{d}$, so $x = y$ and $d \equiv 0 \pmod{4d}$, which is impossible; hence $S_a \cap (S_a - d) = \emptyset$. Finally, if $a_k = c + [k \geq r]$ then $S_a = \{-dc + k : k < r\} \cup \{-dc - d + k : r \leq k < d\} = -dc + [r - d, r)$. Conversely an interval of length d is a transversal, so it is S_a for a unique a ; the $4d$ intervals and the $4d$ one-step configurations therefore correspond to each other bijectively. $\square$

Proof of Theorem B from Theorem A. Let $a \in \mathbb{Z}_4^d$. By Lemma 3.1 and Theorem A with $N = 4d$, $A = S_a$, $B = S_a - d$ (so that $|A| = |B| = d$ and $|C| = 2d \geq 1$),

$$F(a) = \frac{1}{2}\langle S_a, S_a - d \rangle - \frac{d}{2} \leq \frac{1}{2}\Phi_{4d}(d, d) - \frac{d}{2} = F(0) = F_{\text{DKZ}},$$

because $S_0 = [0, d)$ and $S_0 - d = [-d, 0)$. Equality holds if and only if $(S_a, S_a - d)$ is a rotation of $([0, d), [-d, 0))$, that is, if and only if S_a is an interval, that is, if and only if a is one-step.

Now let $V_0, \dots, V_{d-1}$ commute, $V_k^4 = 1$, in a von Neumann algebra with faithful normal tracial state τ (for matrices, $\tau = \operatorname{tr}$). The spectral projections $P_{k,s}$ of V_k ($V_k = \sum_{s \in \mathbb{Z}_4} i^{-s} P_{k,s}$) commute, so $\Pi_a = \prod_k P_{k,a_k}$, $a \in \mathbb{Z}_4^d$, is a finite family of orthogonal projections with sum 1 and $V_k = \sum_a i^{-a_k} \Pi_a$. Then $\tau(V_j V_k^*) = \sum_a \tau(\Pi_a) i^{a_k - a_j}$ and

$$F(V) = \sum_{a \in \mathbb{Z}_4^d} \tau(\Pi_a) F(a) \leq F_{\text{DKZ}},$$

since $\tau(\Pi_a) \geq 0$ and $\sum_a \tau(\Pi_a) = 1$. Equality forces $\tau(\Pi_a) = 0$, hence $\Pi_a = 0$, for every a that is not one-step. For matrices this is the statement about joint eigenvectors. $\square$

Proof of Corollary C. Let R have equal links and let V be its reduced family. By Proposition 2.4 the V_k commute, so by Theorems 2.3 and B, $I_d(R) = 4F(V)/(d(d-1)) \leq 4F_{\text{DKZ}}/(d(d-1)) = I_{\text{ME}}(d)$.

Suppose equality holds. Diagonalise the V_k in a common orthonormal basis $(v_\mu)_{\mu=1}^M$ (this conjugates W by $\mathbf{1} \otimes u_0$ and leaves $X \otimes \mathbf{1}$ unchanged). By Theorem B every $a(\mu)$ is one-step, $a_k(\mu) = c_\mu + [k \geq r_\mu]$, and $R^V = \bigoplus_\mu R^{a(\mu)}$. For $0 \leq r \leq d-1$ one computes $X^r W_0 X^{-r} = \operatorname{diag}(z^{(k-r) \bmod d}) = z^{-r} i \operatorname{diag}(z^k i^{-[k \geq r]})$, using $z^d = i$. Hence $W_{a(\mu)} = \gamma_\mu X^{r_\mu} W_0 X^{-r_\mu}$ with $|\gamma_\mu| = 1$, and since X commutes with X^{r_μ} , $R_i^{a(\mu)} = X^{r_\mu} R_i^{(0)} X^{-r_\mu}$ for $i = 1, \dots, 4$. So R^V is unitarily equivalent to $R^{(0)} \otimes \mathbf{1}_M$, and by Theorem 2.3(2) R is unitarily equivalent to the restriction of $R^{(0)} \otimes \mathbf{1}_M$ to an invariant subspace H' . The $*$ -algebra generated by $R_1^{(0)}, \dots, R_4^{(0)}$ is all of $M_d(\mathbb{C})$: it contains X and $R_1^{(0)}(R_2^{(0)})^{-1} = XW_0X^{-1}W_0^* = z^{-1}(\mathbf{1} + (i-1)|0\rangle\langle 0|)$, hence $|0\rangle\langle 0|$ and all matrix units $X^k|0\rangle\langle 0|X^{-l}$. Therefore the commutant of $\{R_i^{(0)} \otimes \mathbf{1}_M\}$ is $\mathbf{1} \otimes M_M(\mathbb{C})$, and the invariant subspace H' , being the range of a projection in this commutant, is $\mathbb{C}^d \otimes K$ for a subspace K . So $D = d \dim K$ and R is unitarily equivalent to $R^{(0)} \otimes \mathbf{1}_K$, hence (Section 2.4) to

DKZ $\otimes \mathbf{1}_K$. A simultaneous conjugation $R_i \mapsto uR_iu^*$ of the unitaries (6) is the same as the local unitary $u \otimes \bar{u}$ applied to the strategy, and $(u \otimes \bar{u})\Phi_D$ is again maximally entangled. Conversely DKZ $\otimes \mathbf{1}$ attains $I_{\text{ME}}(d)$ and has equal links.

For the clock-phase strategies, $I_d(R^a) = 4F(a)/(d(d-1))$, and $F(a) \leq F_{\text{DKZ}}$ with equality exactly for one-step a by the first part of the proof of Theorem B. $\square$

4. PROOF OF THEOREM A

Fix $N \geq 2$, write $\kappa(u) = \cot(\pi u/N)$ also for real $u \notin N\mathbb{Z}$, and recall $\kappa(0) = 0$ on $\mathbb{Z}_N$. The function κ is odd and N -periodic. For $X \subseteq \mathbb{Z}_N$ let

$$p_X(t) = \sum_{s \in X} \kappa(t-s) \quad (t \in \mathbb{Z}_N).$$

Two facts will be used repeatedly: $\langle Y, X \rangle = -\langle X, Y \rangle$ (oddness; in particular $\langle X, X \rangle = 0$), and $\sum_{u \in \mathbb{Z}_N} \kappa(u) = 0$ (the terms u and $-u$ cancel, and $\kappa(N/2) = 0$ if N is even), so that $\langle X, \mathbb{Z}_N \rangle = 0$ for every X .

Given disjoint A, B we always put $C = \mathbb{Z}_N \setminus (A \cup B)$ and $c = |C|$, and think of the elements of A, B, C as coloured by the three colours A, B, C .

Trivial cases. If $a = 0$ or $b = 0$, both $\langle A, B \rangle$ and $\Phi_N(a, b)$ are empty sums. If $c = 0$, then $\langle A, B \rangle = \langle A, \mathbb{Z}_N \rangle - \langle A, A \rangle = 0$, and $\Phi_N(a, b) = 0$ for the same reason. So from now on $a, b, c \geq 1$, and in particular $N \geq 3$.

4.1. Cyclic symmetry.

Lemma 4.1 (cyclic symmetry). *The three pairings are equal, $\langle A, B \rangle = \langle B, C \rangle = \langle C, A \rangle =: Q$. Consequently*

$$Q = \sum_{y \in B} p_C(y) = \sum_{t \in C} p_A(t) = \sum_{x \in A} p_B(x).$$

Proof. $0 = \langle A, \mathbb{Z}_N \rangle = \langle A, A \rangle + \langle A, B \rangle + \langle A, C \rangle$ gives $\langle A, B \rangle = -\langle A, C \rangle = \langle C, A \rangle$, and $0 = \langle B, \mathbb{Z}_N \rangle$ gives $\langle B, C \rangle = -\langle B, A \rangle = \langle A, B \rangle$. The three formulas are $\langle B, C \rangle$, $\langle C, A \rangle$ and $\langle A, B \rangle$ written out. $\square$

4.2. The exchange identity. For $1 \leq u \leq N-2$ let $\delta(u) = \kappa(u) - \kappa(u+1)$. Since $\pi u/N$ and $\pi(u+1)/N$ lie in $(0, \pi)$ and $\cot$ is strictly decreasing there, $\delta(u) > 0$.

Lemma 4.2 (exchange identity). *Let $x \in \mathbb{Z}_N$ with $x \in A$ and $x+1 \in B$, and let (A', B') be obtained by exchanging the two colours ($A' = A \setminus \{x\} \cup \{x+1\}$, $B' = B \setminus \{x+1\} \cup \{x\}$). Then*

$$\langle A', B' \rangle - \langle A, B \rangle = p_C(x) - p_C(x+1) = \sum_{t \in C} \delta(x-t) > 0,$$

where $x-t$ is read in $\{1, \dots, N-2\}$. Likewise, exchanging an adjacent pair coloured (B, C) increases the common value Q of Lemma 4.1 by $\sum_{s \in A} \delta(x-s) > 0$, and exchanging an adjacent pair coloured (C, A) increases it by $\sum_{y \in B} \delta(x-y) > 0$.

Proof. The exchange does not change C , so by Lemma 4.1 (applied before and after) $\langle A', B' \rangle - \langle A, B \rangle = \sum_{y \in B'} p_C(y) - \sum_{y \in B} p_C(y) = p_C(x) - p_C(x+1)$. For $t \in C$ we have $t \notin \{x, x+1\}$, so $x-t \bmod N \in \{1, \dots, N-2\}$ and $\kappa(x-t) - \kappa(x+1-t) = \delta(x-t)$. The sum is positive because $C \neq \emptyset$. For a pair coloured (B, C) the set A is unchanged and $Q = \sum_{t \in C} p_A(t)$; for a pair coloured (C, A) the set B is unchanged and $Q = \sum_{x \in A} p_B(x)$. The same computation applies, and $A, B \neq \emptyset$. $\square$

We call a colouring *cyclically ordered* if no adjacent pair $(x, x+1)$ is coloured (A, B) , (B, C) or (C, A) .

Corollary 4.3. *Let $a, b, c \geq 1$. Every maximiser of $\langle A, B \rangle$ (over disjoint A, B of sizes a, b) is cyclically ordered. A cyclically ordered colouring consists of $3w$ nonempty maximal blocks of consecutive points, for some integer $w \geq 1$ (the winding number), which occur in the cyclic order*

$$C_1, B_1, A_1, C_2, B_2, A_2, \dots, C_w, B_w, A_w$$

in the direction of increasing x . The colourings with $w = 1$ are exactly the rotations of $([0, a), [-b, 0))$.

Proof. The first assertion follows from Lemma 4.2, since exchanges preserve the sizes. In a cyclically ordered colouring every colour change from x to $x + 1$ is of type $B \rightarrow A$, $A \rightarrow C$ or $C \rightarrow B$; assigning $C \mapsto 0$, $B \mapsto 1$, $A \mapsto 2$ in $\mathbb{Z}_3$, each change adds 1. Going once around the circle the total change is 0 in $\mathbb{Z}_3$, so the number of changes is $3w$ for an integer w , and $w \geq 1$ because all three colours occur. The maximal blocks between consecutive changes then follow the stated order. For $w = 1$ there is one block of each colour, in the order C, B, A ; so $B = r + [-b, 0)$ and $A = r + [0, a)$ for some r . $\square$

4.3. Smearing. Let $\mathbb{T}_N = \mathbb{R}/N\mathbb{Z}$ with Lebesgue measure. For $X \subseteq \mathbb{Z}_N$ let $\tilde{X} = \bigcup_{x \in X} [x, x + 1) \subseteq \mathbb{T}_N$, and for measurable $U, V \subseteq \mathbb{T}_N$ put

$$Q_c(U, V) = \int_U \int_V \kappa(\xi - \eta) d\eta d\xi$$

whenever the integral converges absolutely. For $m \in \{1, \dots, N - 1\}$ define

$$\bar{\kappa}(m) = \int_0^1 \int_0^1 \kappa(m + s - t) ds dt = \int_{-1}^1 (1 - |v|) \kappa(m + v) dv, \quad e(m) = \kappa(m) - \bar{\kappa}(m).$$

The integrand of $\bar{\kappa}(m)$ is bounded: the only possible poles, at $v = -1$ for $m = 1$ and at $v = 1$ for $m = N - 1$, sit where the weight $1 - |v|$ vanishes to first order.

Lemma 4.4 (smearing). *For disjoint $A, B \subseteq \mathbb{Z}_N$ the integral $Q_c(\tilde{A}, \tilde{B})$ converges absolutely and*

$$\langle A, B \rangle = Q_c(\tilde{A}, \tilde{B}) + E(A, B), \quad E(A, B) = \sum_{x \in A, y \in B} e(x - y),$$

where $x - y$ is read in $\{1, \dots, N - 1\}$.

Proof. $Q_c(\tilde{A}, \tilde{B}) = \sum_{x \in A, y \in B} \int_0^1 \int_0^1 \kappa(x - y + s - t) ds dt$. For $x - y \equiv m$ the integrand is singular only if $m + s - t \in N\mathbb{Z}$, which for $s, t \in [0, 1]$ happens only at a corner of the square ($m = 1, s = 0, t = 1$, or $m = N - 1, s = 1, t = 0$). There $|\kappa|$ is bounded by a constant times $1/(s + 1 - t)$, respectively $1/(1 - s + t)$, which is integrable on the square. The substitution $v = s - t$, whose density on $[-1, 1]$ is $1 - |v|$, gives the second expression for $\bar{\kappa}$, and the cell integral equals $\bar{\kappa}(x - y)$. $\square$

4.4. The continuous inequality. Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. A finite union V' of pairwise disjoint open arcs (p_j, q_j) , $j = 1, \dots, n$, with $p_j < q_j < p_j + 2\pi$ and total length $\beta = \sum_j (q_j - p_j) \in (0, 2\pi)$, has the conjugate function

$$(11) \quad g_{V'}(\theta) = \frac{1}{2\pi} \text{p.v.} \int_{V'} \cot \frac{\theta - \varphi}{2} d\varphi = \frac{1}{\pi} \sum_{j=1}^n \log \left| \frac{\sin \frac{\theta - p_j}{2}}{\sin \frac{\theta - q_j}{2}} \right| \quad (\theta \notin \{p_j, q_j\}),$$

since $\frac{d}{d\varphi} (-2 \log |\sin \frac{\theta - \varphi}{2}|) = \cot \frac{\theta - \varphi}{2}$. It has logarithmic singularities at the endpoints and is integrable on $\mathbb{T}$.

Lemma 4.5 (distribution of the conjugate function). *In this situation, for every $\lambda \in \mathbb{R}$,*

$$|\{\theta \in \mathbb{T} \setminus V' : g_{V'}(\theta) > \lambda\}| = 2\pi \mu_\beta((0, e^{-\pi\lambda})),$$

where μ_β is the Cauchy distribution on $\mathbb{R}$ with density $\frac{1}{\pi} \frac{\sin(\beta/2)}{(t - \cos(\beta/2))^2 + \sin^2(\beta/2)}$. In particular the left side depends only on β and λ .

Proof. Let $\mathbb{D}$ be the open unit disc and $\mathbb{H}$ the open upper half plane. Consider the rational function

$$\Psi(z) = e^{-i\beta/2} \prod_{j=1}^n \frac{e^{iq_j} - z}{e^{ip_j} - z}.$$

Ψ maps $\mathbb{D}$ into $\mathbb{H}$. Let $\mathcal{F}(z) = \frac{1}{2\pi} \int_{V'} \frac{e^{i\varphi} + z}{e^{i\varphi} - z} d\varphi$ ($z \in \mathbb{D}$), an analytic function whose real part is the Poisson integral of $\mathbf{1}_{V'}$. As $0 < |V'| < 2\pi$, the positivity of the Poisson kernel gives $0 < \operatorname{Re} \mathcal{F} < 1$ on $\mathbb{D}$. For one arc (p, q) , the partial fractions $\frac{\zeta + z}{\zeta(\zeta - z)} = \frac{2}{\zeta - z} - \frac{1}{\zeta}$ and the substitution $\zeta = e^{i\varphi}$ give $\frac{1}{2\pi} \int_p^q \frac{e^{i\varphi} + z}{e^{i\varphi} - z} d\varphi = \frac{1}{\pi i} \log \frac{e^{iq} - z}{e^{ip} - z} - \frac{q-p}{2\pi}$ for some branch of the logarithm. Summing over the arcs and exponentiating, $\exp(i\pi \mathcal{F}) = \Psi$ on $\mathbb{D}$, whatever the branches. Hence $\arg \Psi = \pi \operatorname{Re} \mathcal{F} \in (0, \pi)$, that is, $\Psi(\mathbb{D}) \subseteq \mathbb{H}$. Also $\Psi(0) = e^{-i\beta/2} e^{i\beta} = e^{i\beta/2}$.

Boundary values. For $\theta \notin \{p_j, q_j\}$, $e^{iq_j} - e^{i\theta} = -2ie^{i(q_j+\theta)/2} \sin \frac{\theta - q_j}{2}$ and similarly for p_j , so

$$\Psi(e^{i\theta}) = \prod_{j=1}^n \frac{\sin \frac{\theta - q_j}{2}}{\sin \frac{\theta - p_j}{2}} \in \mathbb{R} \setminus \{0\}, \quad |\Psi(e^{i\theta})| = e^{-\pi g_{V'}(\theta)}.$$

Each factor is 2π -periodic in θ ; taking $\theta \in (p_j, p_j + 2\pi)$, its denominator is positive and its numerator is negative exactly for $\theta < q_j$. So the j -th factor is negative exactly on the j -th arc, and, the arcs being disjoint, $\Psi(e^{i\theta}) < 0$ if and only if $\theta \in V'$. Therefore

$$\{\theta \in \mathbb{T} \setminus V' : g_{V'}(\theta) > \lambda\} = \{\theta : 0 < \Psi(e^{i\theta}) < e^{-\pi\lambda}\}$$

up to finitely many points.

The law of the boundary values. Let $w_0 = \Psi(0) = e^{i\beta/2} \in \mathbb{H}$ and $\mathcal{M}(\zeta) = (\zeta - w_0)/(\zeta - \bar{w}_0)$, a Möbius map of $\mathbb{H}$ onto $\mathbb{D}$ and of $\mathbb{R} \cup \{\infty\}$ onto the unit circle, with $\mathcal{M}(w_0) = 0$. The rational function $B = \mathcal{M} \circ \Psi$ maps $\mathbb{D}$ into $\mathbb{D}$; it has no poles on the closed disc (it could only have poles where $\Psi = \bar{w}_0$, which does not happen there; at the poles e^{ip_j} of Ψ it takes the value 1), and $|B| = 1$ on the unit circle. For $k \geq 1$ the mean value property of B^k gives $\frac{1}{2\pi} \int_0^{2\pi} B(e^{i\theta})^k d\theta = B(0)^k = 0$, and by conjugation also $\frac{1}{2\pi} \int \bar{B}^k = 0$. So the image of the normalised arc length under $\theta \mapsto B(e^{i\theta})$ has the Fourier coefficients of the normalised arc length, and is therefore the normalised arc length. Finally, $\Psi(e^{i\theta}) = \mathcal{M}^{-1}(B(e^{i\theta}))$ off finitely many points, and $\mathcal{M}^{-1}$ carries the normalised arc length to the harmonic measure of $\mathbb{H}$ at $\mathcal{M}^{-1}(0) = w_0$, which is the Cauchy distribution with centre $\operatorname{Re} w_0 = \cos(\beta/2)$ and scale $\operatorname{Im} w_0 = \sin(\beta/2)$. (For a bounded continuous f on $\mathbb{R}$ with a limit at ∞ , its Poisson extension u to $\mathbb{H}$ satisfies $u(w_0) = \frac{1}{2\pi} \int f(\mathcal{M}^{-1}(e^{it})) dt$ by the mean value property of the harmonic function $u \circ \mathcal{M}^{-1}$ at 0.) So the image of $d\theta/2\pi$ under $\theta \mapsto \Psi(e^{i\theta})$ is μ_β , and the lemma follows. $\square$

For sets E of finite measure on the line, the fact that the distribution function of the Hilbert transform of $\mathbf{1}_E$ depends only on $|E|$ is a theorem of Stein and Weiss [21]; for sums of point masses the analogous statement is Boole's identity [3] (see also Loomis [16] and [6]). That the distributions on E and on its complement depend only on $|E|$ separately is due to Laeng [13] (see the discussion in [8, Section 4]). Lemma 4.5 is the circle version of the latter statement for finite unions of arcs, which is all we need; the proof above is the standard one, made explicit by

the fact that Ψ is a rational function. Proposition 4.6 below is then a direct consequence of the bathtub principle, and we do not claim novelty for it.

Proposition 4.6 (continuous rearrangement). *Let $U, V \subseteq \mathbb{T}_N$ be disjoint, each a finite union of half-open intervals, with $|U| = a$ and $|V| = b$, where $0 < b < N$. Then $Q_c(U, V)$ converges absolutely and*

$$Q_c(U, V) \leq Q_c([0, a), [-b, 0)).$$

Proof. Absolute convergence: the closures of U and V meet in finitely many points. Near such a point $|\kappa(\xi - \eta)|$ is at most a constant times $1/|\xi - \eta|$, with ξ and η on opposite sides of the point, and $\int_0^1 \int_0^1 \frac{ds dt}{s+t} < \infty$; elsewhere the integrand is bounded.

Rescale by $\theta = 2\pi\xi/N$, $\varphi = 2\pi\eta/N$, and let $U', V' \subseteq \mathbb{T}$ be the images of U, V ; after merging touching intervals and discarding finitely many points, V' is a union of disjoint open arcs of total length $\beta = 2\pi b/N \in (0, 2\pi)$, and $U' \subseteq \mathbb{T} \setminus V'$ has measure $\alpha = 2\pi a/N$. By Fubini and (11),

$$Q_c(U, V) = \left(\frac{N}{2\pi}\right)^2 \int_{U'} \int_{V'} \cot \frac{\theta - \varphi}{2} d\varphi d\theta = \frac{N^2}{2\pi} \int_{U'} g_{V'}(\theta) d\theta.$$

Let $\Omega = \mathbb{T} \setminus V'$ and let t_0 be such that the set $U^* = \{\theta \in \Omega : g_{V'}(\theta) > t_0\}$ has measure α ; such t_0 exists because $g_{V'}$ is real analytic and non-constant on each arc of Ω , so its level sets are finite and the distribution function is continuous. Then $g_{V'} \leq t_0$ on $U' \setminus U^*$, $g_{V'} \geq t_0$ on $U^* \setminus U'$, and $|U' \setminus U^*| = |U^* \setminus U'|$, so

$$\int_{U'} g_{V'} - \int_{U^*} g_{V'} = \int_{U' \setminus U^*} g_{V'} - \int_{U^* \setminus U'} g_{V'} \leq 0$$

(the bathtub principle, cf. [15, Thm. 1.14]). The value $\int_{U^*} g_{V'} = \int_0^\alpha g^*(s) ds$, with g^* the decreasing rearrangement of $g_{V'}|_\Omega$, depends only on the distribution function of $g_{V'}$ on Ω , hence, by Lemma 4.5, only on α and β . For the arc $V'_0 = (-\beta, 0)$,

$$g_{V'_0}(\theta) = \frac{1}{\pi} \log \frac{\sin \frac{\theta+\beta}{2}}{\sin \frac{\theta}{2}}, \quad g'_{V'_0}(\theta) = \frac{1}{2\pi} \left[\cot \frac{\theta+\beta}{2} - \cot \frac{\theta}{2} \right] < 0 \quad (0 < \theta < 2\pi - \beta),$$

so its superlevel sets in $\mathbb{T} \setminus V'_0 = (0, 2\pi - \beta)$ are initial intervals and $U^* = (0, \alpha)$. Hence $\int_{U'} g_{V'} \leq \int_0^\alpha g_{V'_0}$, which after rescaling is the claim. $\square$

Remark 4.7 (degeneracy of the continuous problem). The continuous inequality has many maximisers. For an integer $w \geq 1$ let $\pi_w(\theta) = w\theta$ on $\mathbb{T}$. The conjugate function commutes with $f \mapsto f \circ \pi_w$ (the multiplier $-i \operatorname{sgn}(n)$ is invariant under $n \mapsto wn$), and π_w preserves the normalised measure, so the w arcs $\pi_w^{-1}((-\beta, 0))$ and the w arcs $\pi_w^{-1}((0, \alpha))$ give the same value $\int_0^\alpha g_{V'_0}$. In the discrete problem these correspond to cyclically ordered colourings with winding number w , which are therefore excluded only by the junction terms $E(A, B)$ of Lemma 4.4.

Remark 4.8 (the discrete problem needs more). (i) The discrete analogue of Lemma 4.5 fails: for $N = 7$ the values of p_B on $\mathbb{Z}_7 \setminus B$ are $\pm 2.874, \pm 1.026, 0$ for $B = \{0, 1\}$, but $\pm 2.305, \pm 0.569, 0$ for $B = \{0, 2\}$ (to three decimals). This is why the proof passes through the smeared sets and pays for the discretisation separately. (ii) Theorem A is special to the cotangent kernel. For $N = 20$ and the odd kernel $\kappa' = \mathbf{1}_{\{1\}} - \mathbf{1}_{\{-1\}}$, which is also non-increasing on $\{1, \dots, N-1\}$, the pair $A = \{0, 10\}$, $B = \{9, 19\}$ gives $\sum_{x \in A, y \in B} \kappa'(x - y) = 2$, while two adjacent intervals of sizes 2 and 2 give 1.

4.5. The junction function. For real $u \geq 1$ let

$$\varepsilon(u) = \frac{1}{u} - \int_{-1}^1 \frac{1-|v|}{u+v} dv = \frac{1}{u} - [(u+1)\log(u+1) - 2u\log u + (u-1)\log(u-1)]$$

(with $0 \log 0 = 0$), and $\varepsilon(-u) = -\varepsilon(u)$. The second expression holds because $\int_{-1}^1 (1-|v|)f(u+v) dv$ is the second difference $G(u+1) - 2G(u) + G(u-1)$ of any G with $G'' = f$, and $G(x) = x \log x$ for $f(x) = 1/x$ (at $u = 1$ by continuity). In particular $\varepsilon(1) = 1 - 2 \log 2$.

Lemma 4.9 (junction function). (1) For $u \geq 1$, $\varepsilon(u) = -\sum_{i \geq 2} \frac{u^{-(2i-1)}}{i(2i-1)}$. Hence $\varepsilon(u) < 0$,

$|\varepsilon|$ is strictly decreasing on $[1, \infty)$, and for $u \geq 2$

$$|\varepsilon(u)| \leq \frac{1}{6u^3} + \frac{4}{45u^5}.$$

(2) For $m \in \{1, \dots, N-1\}$, $e(m) = \frac{N}{\pi} \sum_{j \in \mathbb{Z}} \varepsilon(m+jN)$, the series converging absolutely.

(3) $e(N-m) = -e(m)$. For $1 \leq m \leq N/2$, $\frac{N}{\pi} \varepsilon(m) \leq e(m) \leq 0$, so $|e(m)| \leq \frac{N}{\pi} |\varepsilon(m)|$. Moreover $|e(1)| \geq \frac{N}{\pi} (|\varepsilon(1)| - |\varepsilon(N-1)|)$, and for $N \geq 6$

$$|e(1)| \geq 0.38493 \frac{N}{\pi}.$$

(4) For $N \geq 4$ put $\text{sc} = \sum_{m=2}^{\lfloor N/2 \rfloor} (m-1)|e(m)|$ and $\text{fw} = \sum_{m=2}^{\lfloor N/2 \rfloor} m|e(m)|$. Then

$$\begin{aligned} \text{sc} &\leq \left[\frac{\zeta(2) - \zeta(3)}{6} + \frac{4}{45} (\zeta(4) - \zeta(5)) \right] \frac{N}{\pi} \leq 0.07785 \frac{N}{\pi}, \\ \text{fw} &\leq \left[\frac{\zeta(2) - 1}{6} + \frac{4}{45} (\zeta(4) - 1) \right] \frac{N}{\pi} \leq 0.11481 \frac{N}{\pi}. \end{aligned}$$

Proof. (1) For $u > 1$ and $|v| \leq 1$, $\frac{1}{u+v} = \sum_{n \geq 0} (-v)^n u^{-n-1}$ uniformly in v . Odd powers integrate to zero against $1-|v|$, and $\int_{-1}^1 (1-|v|)v^{2i} dv = \frac{2}{(2i+1)(2i+2)}$, so $\int_{-1}^1 \frac{1-|v|}{u+v} dv = \sum_{i \geq 0} \frac{u^{-2i-1}}{(i+1)(2i+1)}$. The term $i = 0$ is $1/u$; shifting the index gives the series for $\varepsilon(u)$. At $u = 1$ it follows by monotone convergence as $u \downarrow 1$ (all terms are positive and both sides are continuous). Each term of $|\varepsilon(u)|$ is strictly decreasing in u . For $u \geq 2$ the term $i = 2$ is $1/(6u^3)$, and for $i \geq 3$ we have $i(2i-1) \geq 15$ and $u^{-(2i-1)} \leq u^{-5} 4^{-(i-3)}$, so the remaining terms add up to at most $\frac{u^{-5}}{15} \sum_{j \geq 0} 4^{-j} = \frac{4}{45} u^{-5}$. (The exact value of $\sum_{i \geq 3} 4^{3-i}/(i(2i-1))$ is $0.07727 \dots < 0.0773 < 4/45$.)

(2) The partial fraction expansion $\pi \cot \pi s = \lim_{J \rightarrow \infty} \sum_{j=-J}^J \frac{1}{s+j}$, locally uniform on $\mathbb{R} \setminus \mathbb{Z}$, gives $\kappa(t) = \frac{N}{\pi} \lim_J \sum_{|j| \leq J} \frac{1}{t+jN}$. On $[m-1, m+1]$ a term can have a pole only at an endpoint (the term $j = 0$ at $t = 0$ if $m = 1$, the term $j = -1$ at $t = N$ if $m = N-1$); such terms are integrable against the tent weight, which vanishes at the endpoints, and the sum of the remaining terms converges uniformly on $[m-1, m+1]$. Integrating term by term gives $\frac{\pi}{N} e(m) = \lim_J \sum_{|j| \leq J} \varepsilon(m+jN)$, because for every real u with $|u| \geq 1$ one has $\frac{1}{u} - \int_{-1}^1 \frac{1-|v|}{u+v} dv = \varepsilon(u)$ (for $u \leq -1$ substitute $v \mapsto -v$ and use that ε is odd). The series converges absolutely by (1), since $|\varepsilon(u)| \leq \frac{1}{6|u|^3} + \frac{4}{45|u|^5}$ for $|u| \geq 2$.

(3) Both κ and $\bar{\kappa}$ are odd under $m \mapsto N-m$ (for $\bar{\kappa}$ substitute $v \mapsto -v$). Let $1 \leq m \leq N/2$. Grouping the terms of (2) as $j = 0, -1, 1, -2, 2, \dots$ gives

$$\frac{\pi}{N} e(m) = -|\varepsilon(m)| + |\varepsilon(N-m)| - |\varepsilon(N+m)| + |\varepsilon(2N-m)| - \dots,$$

an alternating series whose arguments $m \leq N-m \leq N+m \leq 2N-m \leq \dots$ are non-decreasing, so its terms are non-increasing in absolute value and tend to zero. Hence the sum lies between

the first two partial sums: $-\varepsilon(m) \leq \frac{\pi}{N}e(m) \leq -\varepsilon(m) + |\varepsilon(N-m)| \leq 0$. For $m = 1$ the right inequality gives the lower bound for $|e(1)|$. Numerically $|\varepsilon(1)| = 2 \log 2 - 1 = 0.3862943\dots$ and, for $N \geq 6$, $|\varepsilon(N-1)| \leq |\varepsilon(5)| = \frac{1}{5} - (6 \log 6 - 10 \log 5 + 4 \log 4) = 0.0013551\dots$, so $|\varepsilon(1)| - |\varepsilon(N-1)| \geq 0.3849392\dots \geq 0.38493$.

(4) By (3) and (1), $\text{sc} \leq \frac{N}{\pi} \sum_{m \geq 2} (m-1) \left(\frac{1}{6m^3} + \frac{4}{45m^5} \right)$ and $\text{fw} \leq \frac{N}{\pi} \sum_{m \geq 2} m \left(\frac{1}{6m^3} + \frac{4}{45m^5} \right)$; the sums are the stated combinations of $\zeta(2) = \pi^2/6$, $\zeta(3)$, $\zeta(4) = \pi^4/90$, $\zeta(5)$, whose values are $0.0778480\dots$ and $0.1148066\dots$. $\square$

The numerical constants are rounded in the safe direction; they are re-derived in interval arithmetic by the script `check_constants.py` (Section 5). The proof only needs $|e(1)| > 2\text{sc} + \text{fw}$, and the bounds give $0.38493 > 2 \cdot 0.07785 + 0.11481 = 0.27051$ (in units of N/π); see (12).

4.6. The junction inequality.

Lemma 4.10 (junction inequality). *Let $a, b, c \geq 1$ and let (A, B) be cyclically ordered with winding number $w \geq 2$. Put $A_0 = [0, a)$, $B_0 = [-b, 0)$. Then $N \geq 6$ and*

$$(12) \quad E(A, B) - E(A_0, B_0) \leq -(w-1)|e(1)| + w \text{sc} + \text{fw} \leq \frac{N}{\pi} (0.49974 - 0.30708 w) < 0.$$

Proof. Since there are $3w \geq 6$ nonempty blocks, $N \geq 6$. Sort the pairs $(x, y) \in A \times B$ by $m = x - y \in \{1, \dots, N-1\}$.

$m = 1$: then $y \in B$ and $y + 1 = x \in A$, a colour change $B \rightarrow A$; in a cyclically ordered colouring these are exactly the w boundaries between B_i and A_i . They contribute $-w|e(1)|$.

$m = N-1$: then $x \in A$ and $x + 1 = y \in B$, a change $A \rightarrow B$, which does not occur.

$2 \leq m \leq N/2$: these pairs contribute ≤ 0 by Lemma 4.9(3).

$N/2 < m \leq N-2$: put $m' = N - m = y - x \bmod N \in [2, N/2]$; the pair contributes $e(m) = |e(m')|$. Let $x \in A_i$ and let r be the number of points of A_i after x . Going forward from x one meets these r points, then the nonempty block C_{i+1} , and only then points of B ; so $m' \geq r + 2$. For fixed x and m' there is at most one y . Summing over the points of A_i , for which $r = 0, 1, \dots, |A_i| - 1$, the contribution of A_i is at most

$$\sum_{r \geq 0} \sum_{m'=r+2}^{\lfloor N/2 \rfloor} |e(m')| = \sum_{m'=2}^{\lfloor N/2 \rfloor} (m' - 1) |e(m')| = \text{sc},$$

and the total over the w blocks is at most $w \text{sc}$. Hence $E(A, B) \leq -w|e(1)| + w \text{sc}$.

For the pair (A_0, B_0) all differences $m = x - y$ lie in $[1, a + b - 1] \subseteq [1, N - 2]$, the value $m = 1$ occurs once, and each value m occurs at most m times, because $x \in [0, a)$ and $y = x - m \in [-b, 0)$ force $0 \leq x \leq m - 1$ and y is determined by x . Pairs with $m > N/2$ contribute ≥ 0 and pairs with $2 \leq m \leq N/2$ contribute $\geq -m|e(m)|$ in total for each m . Hence $E(A_0, B_0) \geq -|e(1)| - \text{fw}$. Subtracting gives the first inequality in (12); inserting $|e(1)| \geq 0.38493 N/\pi$, $\text{sc} \leq 0.07785 N/\pi$ and $\text{fw} \leq 0.11481 N/\pi$ from Lemma 4.9 gives $\frac{N}{\pi} (-(w-1)0.38493 + 0.07785 w + 0.11481) = \frac{N}{\pi} (0.49974 - 0.30708 w)$, which is at most $-0.11442 N/\pi < 0$ for $w \geq 2$. $\square$

4.7. Conclusion of the proof.

Proof of Theorem A. The trivial cases were treated at the beginning of this section; let $a, b, c \geq 1$. There are finitely many pairs (A, B) , so a maximiser exists. By Corollary 4.3 it is cyclically ordered with some winding number $w \geq 1$. Suppose $w \geq 2$. By Lemma 4.4, Proposition 4.6 (applied to $U = \tilde{A}$, $V = \tilde{B}$, which are disjoint finite unions of half-open intervals of measures a and b , and $\tilde{A}_0 = [0, a)$, $\tilde{B}_0 = [-b, 0)$) and Lemma 4.10,

$$\langle A, B \rangle = Q_c(\tilde{A}, \tilde{B}) + E(A, B) \leq Q_c(\tilde{A}_0, \tilde{B}_0) + E(A, B) < Q_c(\tilde{A}_0, \tilde{B}_0) + E(A_0, B_0) = \langle A_0, B_0 \rangle,$$

contradicting maximality. So $w = 1$, the maximiser is a rotation of (A_0, B_0) by Corollary 4.3, and the maximum is $\Phi_N(a, b)$ because $\langle \cdot, \cdot \rangle$ is invariant under rotations. Every pair attaining $\Phi_N(a, b)$ is a maximiser, hence a rotation of (A_0, B_0) . $\square$

5. NUMERICAL CROSS-CHECKS

The proofs above are complete without computer assistance. The following checks were run to guard against errors in formulas and signs; they are *numerical* (floating point, except where interval arithmetic is stated) and are not part of the proofs. The scripts and their outputs are in the folders `scripts/` and `logs/` of the accompanying material; `scripts/README.md` lists the commands.

Statement checked	Script	Range and outcome
Theorem B for $M = 1$, by exhaustion	<code>verify_writeup.py</code>	all 4^d configurations, $d = 2, \dots, 9$: $\max F = F_{\text{DKZ}}$ (difference $\leq 3 \cdot 10^{-14}$), exactly $4d$ maximisers, all intervals
Lemma 3.1	<code>verify_writeup.py</code> , <code>cot_rearr.py</code>	identity error $\leq 8 \cdot 10^{-14}$, $d = 2, \dots, 9$
Theorem A by exhaustion	<code>cot_rearr.py</code>	$(N, a, b) = (12, 3, 3), (16, 4, 4), (20, 5, 5), (24, 6, 6), (16, 3, 6), (20, 4, 7)$: maximum $= \Phi_N(a, b)$, attained by adjacent intervals
Lemmas 4.1, 4.2, 4.4	<code>verify_writeup.py</code>	random colourings, $N \in \{8, 12, 20, 33, 40\}$: errors $\leq 4 \cdot 10^{-13}$
Corollary 4.3 (exchanges)	<code>swap_test.py</code>	all colourings for $(N, a, b) = (9, 3, 3), (10, 3, 3), (12, 3, 4), (12, 4, 4)$: 290,710 out-of-order exchanges, none decreases $\langle A, B \rangle$
Lemma 4.9(2)	<code>verify_writeup.py</code>	$N \leq 40$: agreement to $2 \cdot 10^{-7}$ (quadrature accuracy)
Lemma 4.9(3),(4)	<code>verify_junction.py</code>	$N \in \{6, 7, 8, 12, 16, 24, 40, 64, 101\}$: signs and bounds hold; $0.38536 \leq e(1) /(N/\pi) \leq 0.38630$, $sc/(N/\pi) \leq 0.0733$, $fw/(N/\pi) \leq 0.1097$
constants in Lemmas 4.9 and 4.10	<code>check_constants.py</code>	interval arithmetic (60 digits): all stated roundings are valid
whole chain on exchange-flow endpoints	<code>verify_chain.py</code>	random starts, $N \leq 80$: exchange identity to $3 \cdot 10^{-13}$; $Q_c \leq Q_c^{\text{block}}$, $E \leq E^{\text{block}}$ and $\langle A, B \rangle \leq \Phi_N$ at every endpoint
Section 2, Appendix A	<code>check_reduction.py</code>	checks (R1)–(R9) below
Sherali–Adams relaxation	<code>classical_sa.py</code>	$d = 3, \dots, 12$ (Section 6)

The script `check_reduction.py` computes CGLMP values from outcome probabilities with the expression (1) and checks: (R1) Lemma 2.1 on random behaviours; (R2) formula (9) on random projective strategies on Φ_D , $D \leq 6$; (R3) Theorem 2.3(1) for random families V with $M \leq 3$ (non-commuting for $M \geq 2$); (R4) Theorem 2.3(2), by carrying out the twirl and the Stone–von Neumann decomposition of Appendix A on random strategies; (R5) Proposition 2.4 on random strategies (unequal links, non-commuting reduced family) and on equal-link strategies (commuting reduced family); (R6) exhaustion over all clock-phase strategies for $d \leq 7$ (maximum $I_{\text{ME}}(d)$, attained exactly by the $4d$ one-step ones); (R7) the value $I_{\text{ME}}(d)$ of the CGLMP measurements and the identity $R^{\text{DKZ}} = GR^{(0)}G^*$ of Section 2.4, $d \leq 8$; (R8) the projection form (13) and (R9) the Ising form of Section 6, for random non-commuting families. All identities hold to within $2 \cdot 10^{-14}$.

6. DISCUSSION

6.1. An Ising reformulation. A unitary E with $E^4 = 1$ can be written as $E = e^{i\pi/4}(X - iX')/\sqrt{2}$ with commuting involutions X, X' : on the eigenspaces of E for the eigenvalues $1, i, -1, -i$ take $(X, X') = (1, 1), (1, -1), (-1, -1), (-1, 1)$, respectively. Put $E_k = V_k^*$, $X_k = X(E_k)$ and $X_{k+d} = X'(E_k)$ for $0 \leq k \leq d-1$, and extend antiperiodically by $X_{n+2d} = -X_n$. If one expands $\tau(E_j^* E_k)$ and groups the terms by their distance on the ring $\mathbb{Z}_{2d}$, one obtains

$$F(V) = \frac{1}{2} \sum_{n \in \mathbb{Z}_{2d}} \sum_{u=1}^{d-1} \sec \frac{\pi u}{2d} \tau(X_n X_{n+u}),$$

and conversely every family of involutions with $[X_n, X_{n+d}] = 0$ arises. In the commutative sector this is the energy of a classical long-range ferromagnetic Ising ring of $2d$ spins with antiperiodic boundary condition; the DKZ configuration is a single domain wall. Theorem B thus says that for these couplings the single domain wall is the unique ground state (up to translation and global sign). The constraint $[X_n, X_{n+d}] = 0$ matters: the anticommuting choice $X_n = \cos(\frac{\pi n}{2d})\sigma_1 + \sin(\frac{\pi n}{2d})\sigma_2$ (Pauli matrices, $\tau = \frac{1}{2} \text{Tr}$) gives $\tau(X_n X_{n+u}) = \cos \frac{\pi u}{2d}$ and hence the value $d(d-1)$, which exceeds F_{DKZ} because $(d-m) \sec \frac{\pi m}{2d} < d$ for $1 \leq m \leq d-1$ (use $\cos x > 1 - \frac{2x}{\pi}$ on $(0, \frac{\pi}{2})$).

6.2. Why local certificates are not enough. The degeneracy in Remark 4.7 is visible in the clock model itself. Let $w \geq 5$ with $w \equiv 1 \pmod{4}$ divide d , and let $A = \bigcup_{j=0}^{w-1} (\frac{4d}{w}j + [0, \frac{d}{w})) \subseteq \mathbb{Z}_{4d}$. Since $d \equiv \frac{d}{w} \pmod{\frac{4d}{w}}$, the translates $A, A+d, A+2d, A+3d$ are the w -fold unions of the four intervals of length $\frac{d}{w}$ in each period $\frac{4d}{w}$, so they partition $\mathbb{Z}_{4d}$; hence A is a transversal, $A = S_a$ for some $a \in \mathbb{Z}_4^d$, and $(\widetilde{A}, \widetilde{A-d})$ is the w -fold cover of $([0, d], [-d, 0))$. By Remark 4.7, Lemma 4.4 and Lemma 3.1,

$$F_{\text{DKZ}} - F(a) = \frac{1}{2} [E(A_0, B_0) - E(A, A-d)] \quad (A_0 = [0, d], B_0 = [-d, 0)),$$

a difference of junction terms. The pair $(A, A-d)$ is cyclically ordered with winding number w , so by Lemma 4.10 this difference is at least $\frac{2d}{\pi}(0.30708w - 0.49974)$. It is also at most $\frac{2d}{\pi}0.50111w$: all differences occurring in (A_0, B_0) lie in $[1, 2d-1]$, so $E(A_0, B_0) \leq 0$ by Lemma 4.9(3); and in $(A, A-d)$ a point y of a block B_i followed by s further points of B_i has its first A -point at distance $s+1$, so, counting as in the proof of Lemma 4.10, $E(A, A-d) \geq -w \sum_{m=1}^{2d} m|e(m)| = -w(|e(1)| + \text{fw}) \geq -w(0.38630 + 0.11481)\frac{4d}{\pi}$. So the gap is of order d for fixed w , while $F_{\text{DKZ}} \geq d(d-1)/2$: such “ w -fold” configurations are separated from the DKZ configuration only at relative order $1/d$, and one should not expect certificates built from the marginals of a bounded number of sites to resolve this for large d . As an illustration (numerical, linear programming with the HiGHS solver, script `classical_sa.py`): the relaxation of the classical clock model in which one optimises over consistent families of three-site marginals of the differences $a_k - a_j$ is exact for $d = 3, \dots, 8$, but exceeds F_{DKZ} by 0.535, 2.52, 5.23 and 8.53 for $d = 9, 10, 11, 12$. The argument of Section 4 is global: it uses the cyclic order of the whole configuration, and the junction estimate compares different windings directly.

6.3. The non-commutative case. The first steps of the argument have operator versions. Let $V_0, \dots, V_{d-1}$ be arbitrary (not necessarily commuting) unitaries of order dividing four in a tracial von Neumann algebra $(\mathcal{N}, \tau)$, $E_k = V_k^*$, and for $x \in \mathbb{Z}_{4d}$ let Q_x be the spectral projection of E_k for the eigenvalue i^a , where $x = k - da$ ($0 \leq k \leq d-1$, $a \in \mathbb{Z}_4$). Put $A_x = Q_x$, $B_x = Q_{x+d}$ and $N = 4d$. Then $A_x B_x = 0$, $\sum_x A_x = \sum_x B_x = d \cdot 1$, and the computation in the proof of

Lemma 3.1 gives

$$(13) \quad F(V) = \frac{1}{2} \sum_{x,y \in \mathbb{Z}_{4d}} \kappa(x-y) \tau(A_x B_y) - \frac{d}{2}.$$

This suggests the following operator version of Theorem A.

Question 6.1. Let A_x, B_x ($x \in \mathbb{Z}_N$) be orthogonal projections in a tracial von Neumann algebra with $A_x B_x = 0$ for all x , $\sum_x A_x = a \cdot \mathbf{1}$ and $\sum_x B_x = b \cdot \mathbf{1}$. Is $\sum_{x,y} \kappa(x-y) \tau(A_x B_y) \leq \Phi_N(a, b)$?

A positive answer for $N = 4d$ and $a = b = d$ (even only for the families of the form above) would prove the optimality of the DKZ measurements on maximally entangled states for all d . For commuting families the answer is positive by Theorem A, applied to each joint eigenvector. For matrices the same holds if the A_x commute among themselves and the B_x commute among themselves: if (e_i) and (f_j) are joint eigenbases, with $A_x e_i = [x \in S_i] e_i$ and $B_x f_j = [x \in T_j] f_j$, then $|S_i| = a$, $|T_j| = b$, the matrix $|\langle e_i, f_j \rangle|^2$ is doubly stochastic, $\langle e_i, f_j \rangle = 0$ unless $S_i \cap T_j = \emptyset$ (because $A_x B_x = 0$), and the left side equals $\frac{1}{M} \sum_{i,j} |\langle e_i, f_j \rangle|^2 \langle S_i, T_j \rangle \leq \Phi_N(a, b)$. The operator constraints cannot be weakened to $\sum_x \tau(A_x) = a$, $\sum_x \tau(B_x) = b$: a mixture that, with probability $\frac{1}{2}$, uses the sets $([0, 2], [-2, 0])$ and otherwise empty sets has $\sum_x \tau(A_x) = \sum_x \tau(B_x) = 1$ and value $\frac{1}{2} \Phi_{20}(2, 2) = 7.216 \dots > \Phi_{20}(1, 1) = \cot(\pi/20) = 6.313 \dots$ for $N = 20$.

The cyclic symmetry of Lemma 4.1 holds verbatim for such families, with the projections $C_x = \mathbf{1} - A_x - B_x$ and $\langle A, B \rangle_\tau = \sum_{x,y} \kappa(x-y) \tau(A_x B_y)$. The exchange identity of Lemma 4.2 survives in a weak form: if X is a projection with $X \leq A_x$ and $X \leq B_{x+1}$, then replacing $(A_x, B_x, A_{x+1}, B_{x+1})$ by $(A_x - X, B_x + X, A_{x+1} + X, B_{x+1} - X)$ preserves all constraints, leaves every C_t unchanged, and changes the pairing by $\sum_{t \neq x, x+1} \delta(x-t) \tau(X C_t) \geq 0$ (use $\langle A, B \rangle_\tau = \langle B, C \rangle_\tau$). In the non-commutative case such subprojections need not exist, and we do not know operator analogues of Proposition 4.6 and of the junction inequality.

6.4. Summary of the status of Problem 27B (maximally entangled part). The inequality $F(V) \leq F_{\text{DKZ}}$ (equivalently $I_d \leq I_{\text{ME}}(d)$ for projective measurements on maximally entangled states) is proved for commuting families for all d (this paper), and for all families for $d \leq 12$ by exact certificates [17]; for $d = 2$ it is Tsirelson's bound. It is open for non-commuting families when $d \geq 13$. Within the commutative sector, Corollary C also gives the uniqueness ("necessarily of DKZ form") asked for in [10].

APPENDIX A. PROOF OF THEOREM 2.3 AND PROPOSITION 2.4

Write τ for the normalised trace on the relevant matrix algebra.

Invariances. For a quadruple $R = (R_1, R_2, R_3, R_4)$ of unitaries with $R_i^d = 1$ let $\rho(R) = (R_2, R_3, R_4, wR_1)$, again a quadruple of this kind. Directly from the definitions, $L_n^{(0)}(\rho R) = R_2^n R_3^{-n} = L_n^{(1)}(R)$, $L_n^{(1)}(\rho R) = L_n^{(2)}(R)$, $L_n^{(2)}(\rho R) = R_4^n (wR_1)^{-n} = L_n^{(3)}(R)$ and $L_n^{(3)}(\rho R) = w^{-n} (wR_1)^n R_2^{-n} = L_n^{(0)}(R)$. So $L_n^{(i)}(\rho R) = L_n^{(i+1)}(R)$ with the upper index read modulo 4; in particular $\Lambda_n(\rho R) = \Lambda_n(R)$ and $S(\rho R) = S(R)$ by (9). Moreover $\rho^4(R) = wR$ and $\rho^{4d}(R) = R$. For a direct sum $R \oplus R'$ on $\mathbb{C}^D \oplus \mathbb{C}^{D'}$, S is the average of $S(R)$ and $S(R')$ with weights D and D' .

Covariant strategies: proof of (1). Let $R = R^V$ as in (10). From $W^4 = \sum_k |k\rangle\langle k| \otimes w^k = Z_0 \otimes \mathbf{1}$ and $Z_0 X Z_0^* = wX$ we get $W R_4 W^* = W^4 R_1 W^{-4} = wR_1$. Hence $L_n^{(1)} = W L_n^{(0)} W^*$, $L_n^{(2)} = W^2 L_n^{(0)} W^{-2}$, and $W^3 L_n^{(0)} W^{-3} = W^3 R_1^n W^{-3} W^4 R_1^{-n} W^{-4} = w^{-n} R_4^n R_1^{-n} = L_n^{(3)}$. So all four links have the trace $\tau(L_n^{(0)})$. Since $R_1^n W R_1^{-n} = \sum_k |k\rangle\langle k| \otimes W_{k-n}$ with $W_k = z^k V_k$ and

indices modulo d ,

$$L_n^{(0)} = \sum_k |k\rangle\langle k| \otimes W_{k-n} W_k^*, \quad \tau(L_n^{(0)}) = \frac{z^{-n}}{d} \left[\sum_{k=n}^{d-1} \text{tr}(V_{k-n} V_k^*) + i \sum_{k=0}^{n-1} \text{tr}(V_{k-n+d} V_k^*) \right],$$

using $z^d = i$ for $k < n$. Since $w^{-n} = e^{-4i\psi_n}$ and $z^{-n} = e^{-i\psi_n}$, we have $c_n = -1/(1 - e^{-4i\psi_n}) = \frac{i}{2} e^{2i\psi_n} / \sin(2\psi_n)$ and $4c_n z^{-n} = i e^{i\psi_n} / (\sin \psi_n \cos \psi_n) = i \csc \psi_n - \sec \psi_n = -h_n$. Therefore

$$\sum_{n=1}^{d-1} c_n \Lambda_n = 4 \sum_n c_n \tau(L_n^{(0)}) = -\frac{1}{d} \sum_n h_n \left[\sum_{k \geq n} \text{tr}(V_{k-n} V_k^*) + i \sum_{k < n} \text{tr}(V_{k-n+d} V_k^*) \right].$$

The first inner sum runs over the pairs $j = k - n < k$ with $k - j = n$ and gives $\sum_{j < k} h_{k-j} \text{tr}(V_j V_k^*)$. In the second, put $p = k < q = k - n + d$, so $q - p = d - n$; since $\psi_{d-n} = \frac{\pi}{2} - \psi_n$ we have $h_{d-m} = \csc \psi_m - i \sec \psi_m$, i.e. $i h_{d-m} = \overline{h_m}$, and the second part equals $\sum_{p < q} \overline{h_{q-p}} \text{tr}(V_q V_p^*) = \sum_{p < q} \overline{h_{q-p}} \text{tr}(V_p V_q^*)$. Hence $\sum_n c_n \Lambda_n = -\frac{2}{d} F(V)$ and $S(R^V) = 2(d-1) - \frac{2}{d} F(V)$; Lemma 2.1 gives $I_d = 4F(V)/(d(d-1))$.

Twirl: proof of (2). Let R act on $\mathbb{C}^D$ and let $\tilde{R}_i = \bigoplus_{r \in \mathbb{Z}_{4d}} (\rho^r R)_i$ on $\mathbb{C}^{4d} \otimes \mathbb{C}^D$. Then $S(\tilde{R}) = S(R)$, so $I_d(\tilde{R}) = I_d(R)$, and R is the direct summand $r = 0$. Let W be the shift $W(e_{r+1} \otimes v) = e_r \otimes v$. Since $(\rho^{r+1} R)_i = (\rho^r R)_{i+1}$ for $i \leq 3$ and $(\rho^{r+1} R)_4 = w(\rho^r R)_1$, we get $W \tilde{R}_i W^* = \tilde{R}_{i+1}$ ($i \leq 3$), $W \tilde{R}_4 W^* = w \tilde{R}_1$ and $W^{4d} = 1$. The unitary $Z = W^4$ satisfies $Z^d = 1$ and $Z \tilde{R}_1 Z^* = w \tilde{R}_1$. Let $H_k = \ker(Z - w^k)$, $k \in \mathbb{Z}_d$; then $\tilde{R}_1 H_k = H_{k+1}$, all H_k have the same dimension M , and $dM = 4dD$. Choose an orthonormal basis (f_μ) of H_0 and put $e_{k,\mu} = \tilde{R}_1^k f_\mu$, $0 \leq k \leq d-1$. This is an orthonormal basis in which $\tilde{R}_1 = X \otimes \mathbf{1}_M$ (note $\tilde{R}_1^d = 1$) and $Z = Z_0 \otimes \mathbf{1}_M$. Since W commutes with Z , it preserves every H_k , so $W = \sum_k |k\rangle\langle k| \otimes W_k$ with $W_k^4 = w^k$; with $V_k = z^{-k} W_k$ we have $V_k^4 = 1$, and $\tilde{R}$ is exactly R^V in this basis. Another choice of (f_μ) conjugates all V_k by the same unitary. Part (1) now gives $I_d(R) = 4F(V)/(d(d-1))$. The final equivalence in the theorem follows from (1), (2) and $I_{\text{ME}}(d) = 4F_{\text{DKZ}}/(d(d-1))$.

Proof of Proposition 2.4. By the computation above, $L_n^{(i)}(\rho^r R) = L_n^{(i+r)}(R)$ (upper index modulo 4), so $L_n^{(i)}(\tilde{R}) = \bigoplus_r L_n^{(i+r)}(R)$. Hence $\tilde{R}$ has equal links if and only if R has. In the basis $(e_{k,\mu})$, $\tilde{R} = R^V$ and, as shown in the proof of (1), $L_n^{(i)} = W^i L_n^{(0)} W^{-i}$ for all i ; so R^V has equal links if and only if W commutes with every $L_n^{(0)}$. Comparing the diagonal blocks, $W L_n^{(0)} W^* = \sum_k |k\rangle\langle k| \otimes W_k W_{k-n} W_k^* W_k^*$, which equals $L_n^{(0)}$ if and only if $W_k W_{k-n} = W_{k-n} W_k$ for every k . For $n = 1, \dots, d-1$ this says that all W_k , equivalently all V_k , commute pairwise.

DATA AND CODE

The check scripts, their outputs, and a description of the commands are provided with this manuscript (folders `scripts/` and `logs/`). The exact certificates for $d \leq 12$ referred to above are available at <https://github.com/anshm123/CGLMP>.

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